

# Predicting the Exam Performance of Students

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## Introduction

Exam success and general academic performance is often determined by a combination of several habits and behaviors. Factors such as attendance, doing class readings, prior experience, study time, and the amount of sleep a student got the night before can all influence how well someone performs. Understanding how these behaviors relate to scores can provide instructors and future students with insight into learning and help identify which habits contribute most to academic success.

In this project, survey data collected after the first exam was analyzed to determine whether these self-reported metrics could meaningfully predict student outcomes via both classification and regression methods. By examining patterns in the data and applying multiple machine learning algorithms, this analysis aims to evaluate how well these features explain exam performance and where the limitations of the dataset constrain predictive accuracy.

## Background

The dataset consists of survey responses taken after the first exam, asking relevant metrics about class engagement and habits, such as sleep, study time, attendance. Before classification algorithms or regression could be performed, certain pre-processing steps were taken to optimize performance and reduce noise. This consisted of data transformations and feature engineering.

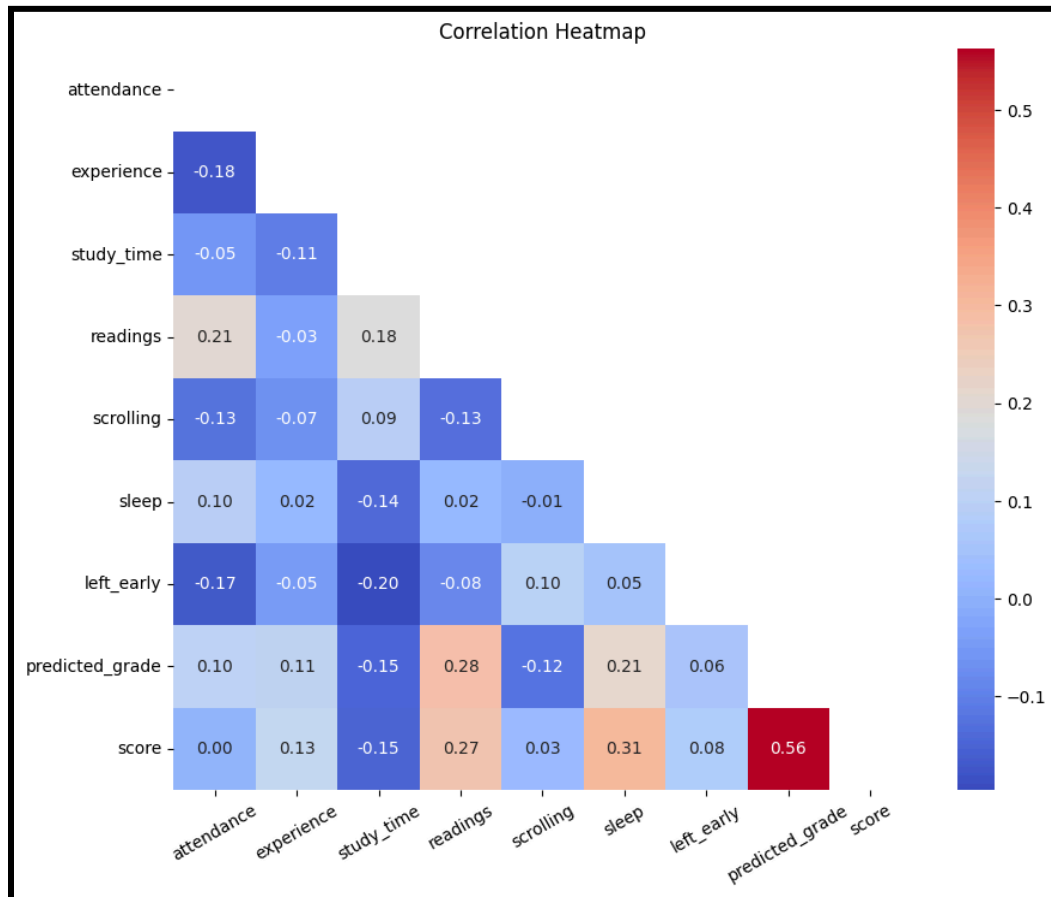
Across all columns, values without information such as “Prefer not to say” or “It’s complicated” were removed. Features with a larger range of values such as sleep, time spent on scrolling apps, and study time were normalized via min-max scaling. Categorical features such as reading, attendance, experience, and predicted grade were converted to numerical values via mapping. Lastly, binning was performed on the student’s raw scores to create target features (pass/fail, letter grades) for the classification algorithms.

Following this, certain fields of the data were dropped to optimize the dataset. Fields not relevant to exam performance, such as maximum points and time of survey completion were dropped. ANOVA tests were performed for year and section columns, to determine whether they played a significant role in exam score.

For section, the test revealed that there was no significant difference between the scores in the 9:30 and 2:00 class, with a p-value of 0.13. Since there were several missing entries and no significance, this justified dropping the section. For student year, there was no significant difference between the scores of Juniors and Seniors, with a p-value close to 1. While sophomores did show significant differences compared to the other two years, they only comprised 20 entries in the column. Due to the high class imbalance towards juniors and lack of significance between juniors and seniors, the year column was also dropped. Across all years, students still have to complete the same pre-requisites and factors such as study time, attendance, and prior experience would outweigh a student’s year.

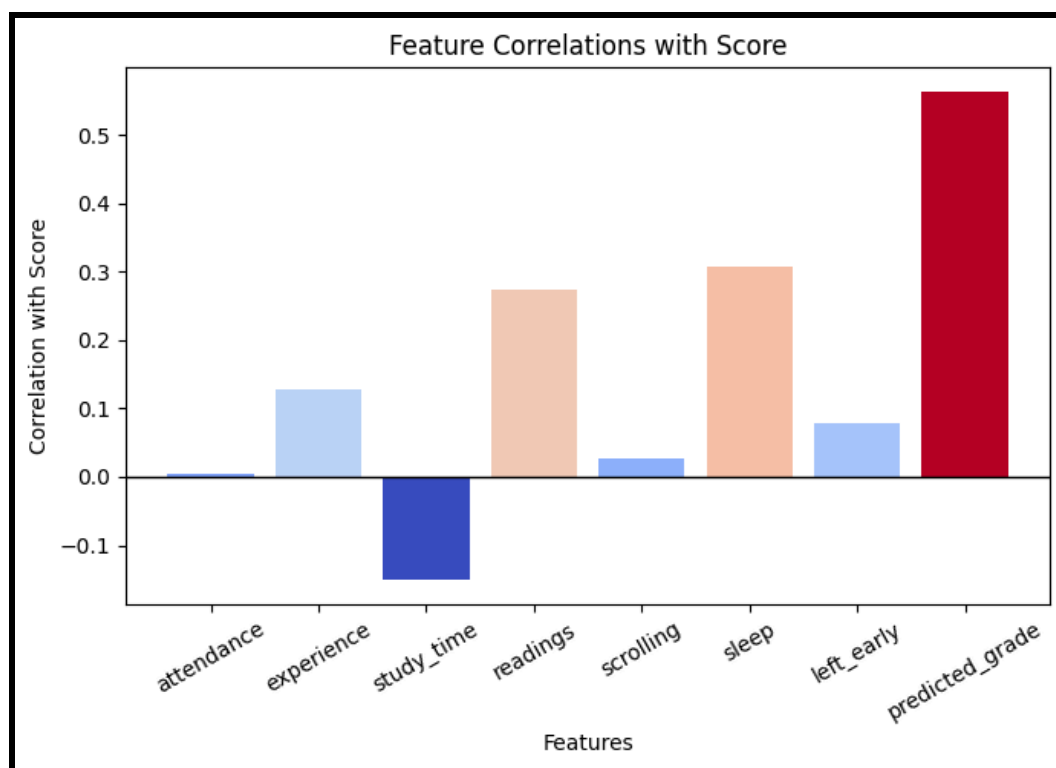
## Data Exploration

After preparing the data for modelling, some data visualization and exploration was done to gain useful insights regarding the correlations. To evaluate any relationships between the features, we examined the data in a correlation heatmap. This visualization allowed for a clear, descriptive summary of interactions between features and correlation with score.



As seen in the blue tiles, the heatmap generally showed that there was low correlation between the variables. Sleep time, readings, and predicted grade seem to have the strongest correlation with score, with readings and sleep showing slight correlation with predicted grade. Since no two features had strong correlations with each other, they were not dropped on this basis.

Correlations between the different features and score were further visualized in a bar plot. This helped visualize which features were adding noise and which ones were considered the most important.



Across both visualizations, predicted grades seemed to have the strongest correlation to the actual exam score, followed by readings and sleep. Since attendance, scrolling, and leaving early displayed near-zero correlation, they were dropped prior to classification. This allowed for a more focused dataset and reduced noise.

## Model Performance and Evaluation

Several machine learning algorithms were used to predict the outcome of students on the exam. Classification algorithms were first used to predict whether a student would pass or fail and then to predict a student's letter grade. A regression algorithm was used to predict a student's numerical score. For the classification algorithms, the features of strongest correlation with score were used: prior experience, study time, readings, sleep, and predicted score. The following metrics were used for the different types of problems:

- Passing/Failing: Accuracy, Precision, Recall
- Letter Grades: Accuracy
- Regression:  $R^2$ , Root Mean Squared Error(RMSE), Mean Absolute Error

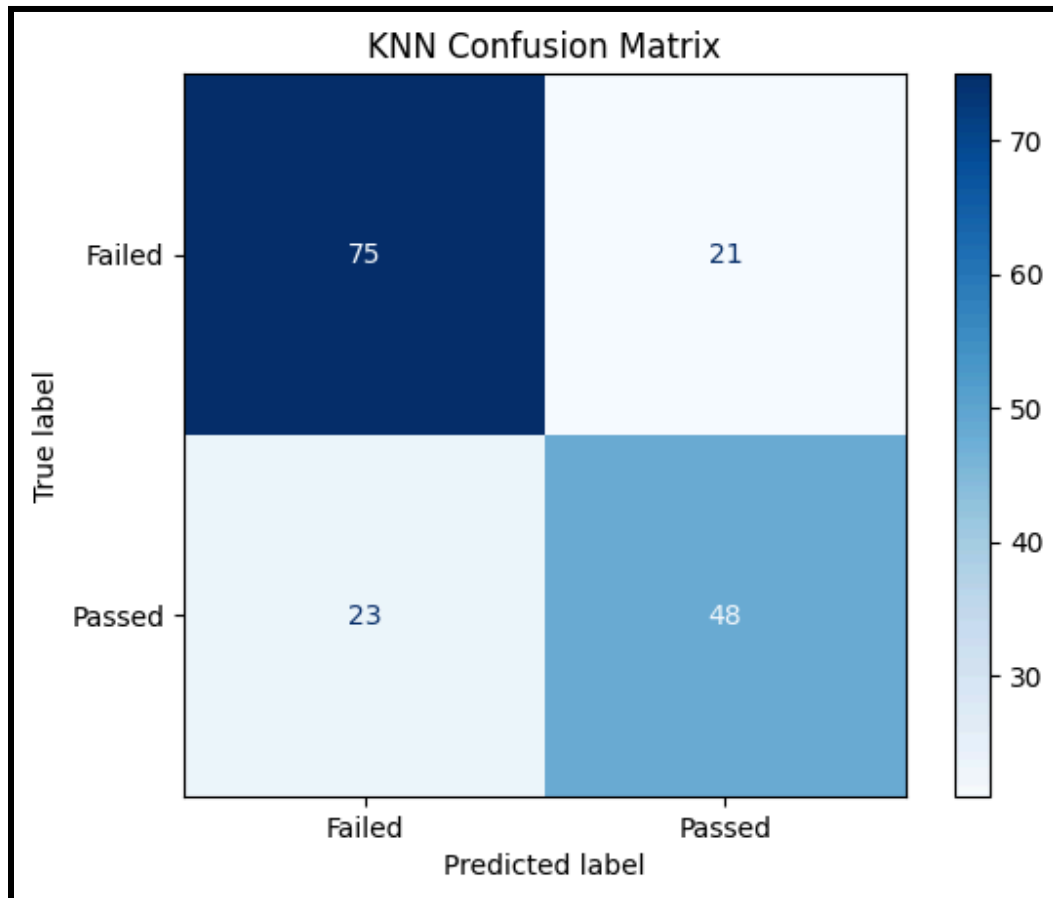
For the first two classification problems, confusion matrices were used to visualize model results, while a residual plot was used for regression.

### Passing vs Failing

Classification algorithms were used to predict whether a student would pass or fail. For this binary classification, K-nearest neighbors, support vector machine(SVM), and a random forest

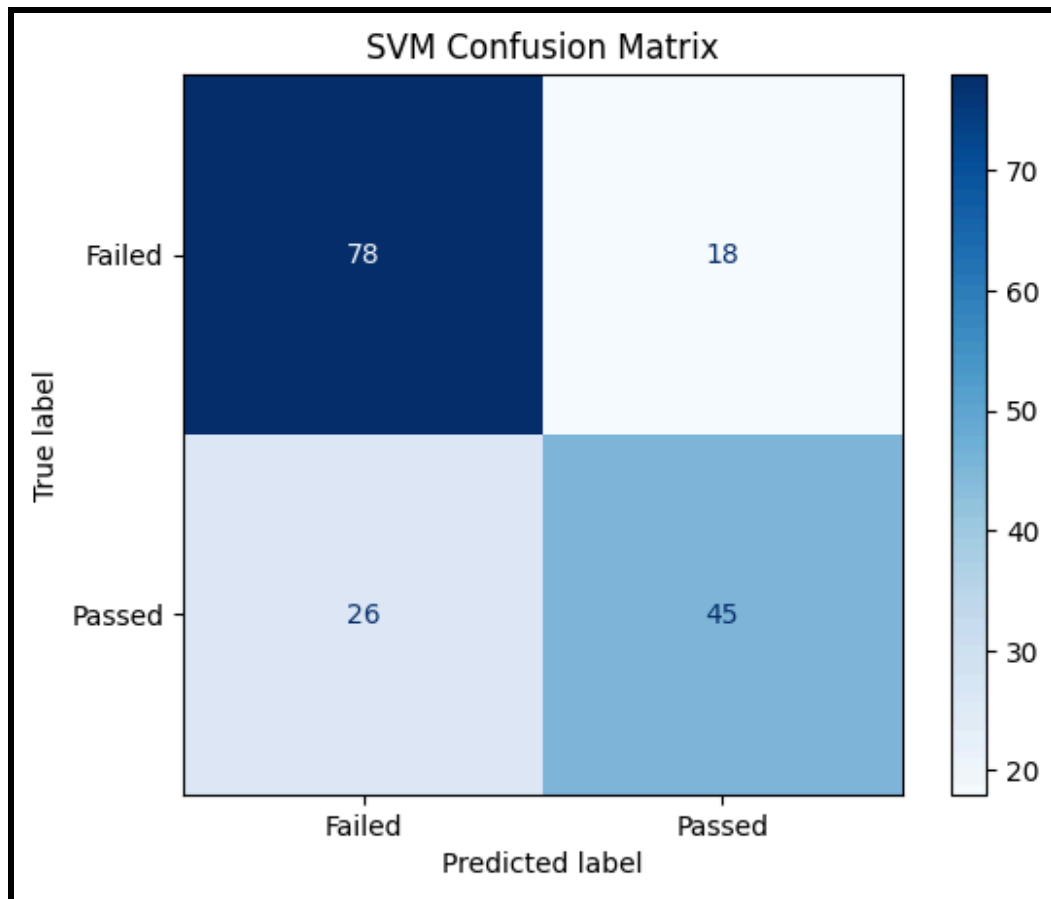
classifier were used. Ten-fold cross validation was used for these algorithms to ensure accurate and reproducible test results.

For K-nearest-neighbors, the K value was initially set to 7 to ensure a good balance of predictive power and generalization. With this value, the model performed with an accuracy of 72.4%. To improve this, a grid search was performed, to look at several different K values as well as manhattan and euclidean distance metrics. Ultimately, the model performed slightly better with  $K = 11$  and euclidean distance, giving an accuracy of 73.6%. The results are shown in the confusion matrix below:



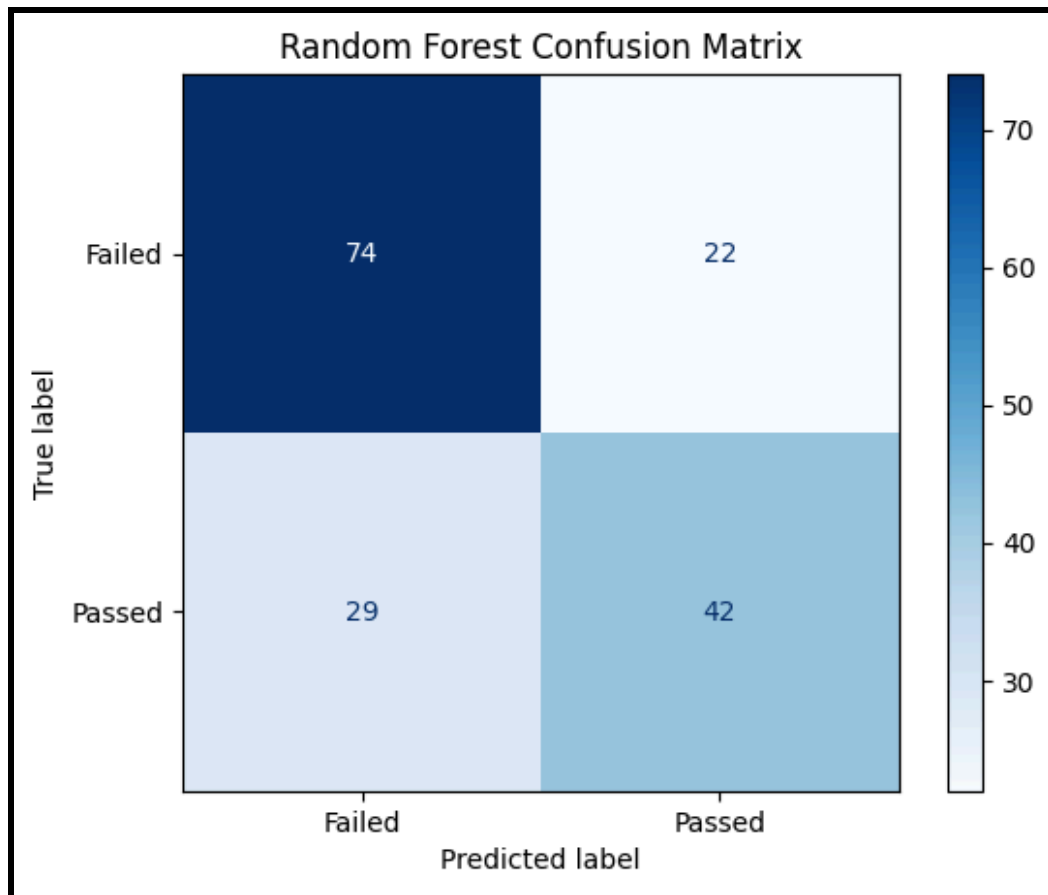
The model performed well for correctly predicting students who failed the exam, labeling 78% of them as correctly failing. However, it struggled more with correctly identifying students who passed the exam, correctly labeling 68% of them as passing. This is likely due to the class imbalance in the training set, since many more students failed the exam than passed. While there were more true negatives than true positives, there were a similar number of false positives, 21, and false negatives, 25.

For SVM, a grid search was performed to choose the best values for gamma, C, and the kernel. For the kernel value, linear was chosen, while the default values of "scale" for gamma and 1 for C were kept. With these parameters, the SVM gave an accuracy of 73.6%. The results were as follows:

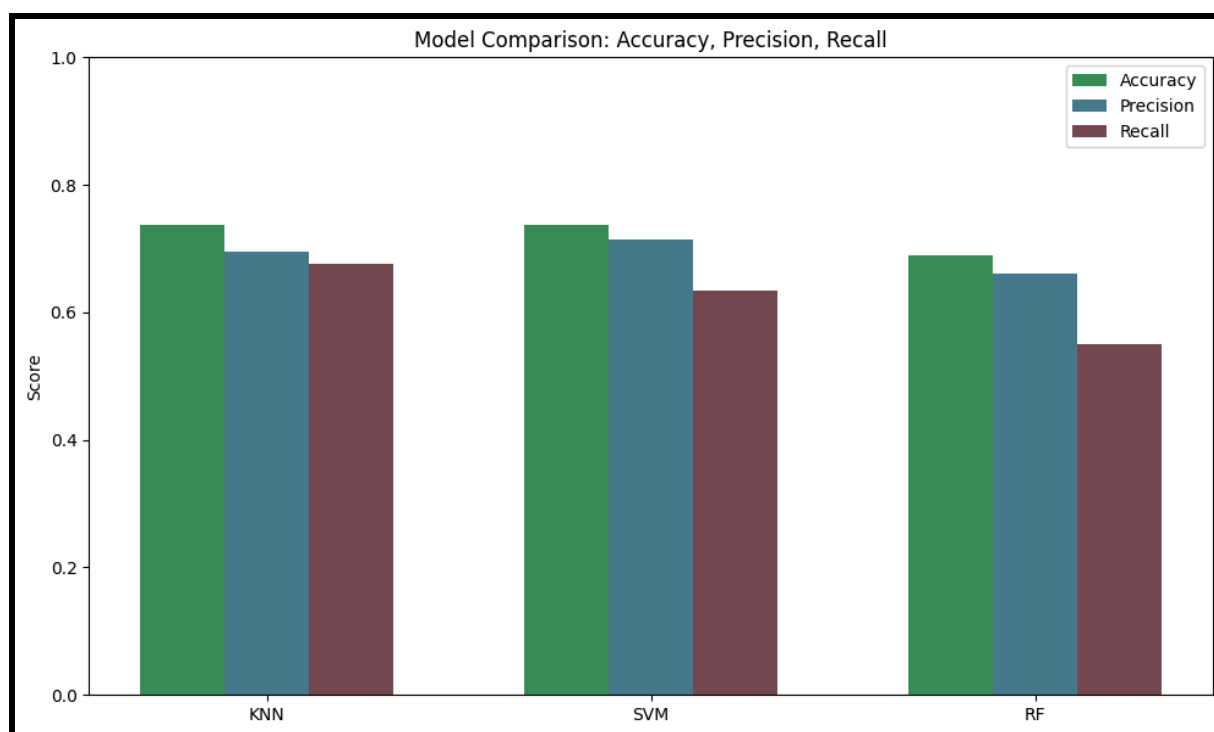


Similar to KNN, the SVM model also performed much better for identifying failed students correctly compared to other categories. Of all students who actually failed, the model correctly identified 81% of them as failing. The model achieved a recall of around 63%, meaning it was labeling many students who actually passed as failing. There were generally more false negatives, 26, compared to 18 false positives.

For the random forest classifier, the hyperparameters chosen based on the grid search were 200 estimators and a max depth of 5. With these parameters, the classifier gave an accuracy of 70%, which was slightly lower than K-Nearest Neighbors and SVM. This is visualized in the confusion matrix below:



The confusion matrix revealed that the model excelled at correctly labeling students who actually failed the exam as failing. However, there were a larger number of false negatives, meaning students who actually passed the exam frequently got labelled as failing. The recall was lower than the previous two, at only 59%. It is likely that the dataset was not large enough and that the features were too weak for the random forest to pick up on. To better compare the performances of models against each other, the precision, recall, and accuracy of these three models were plotted:

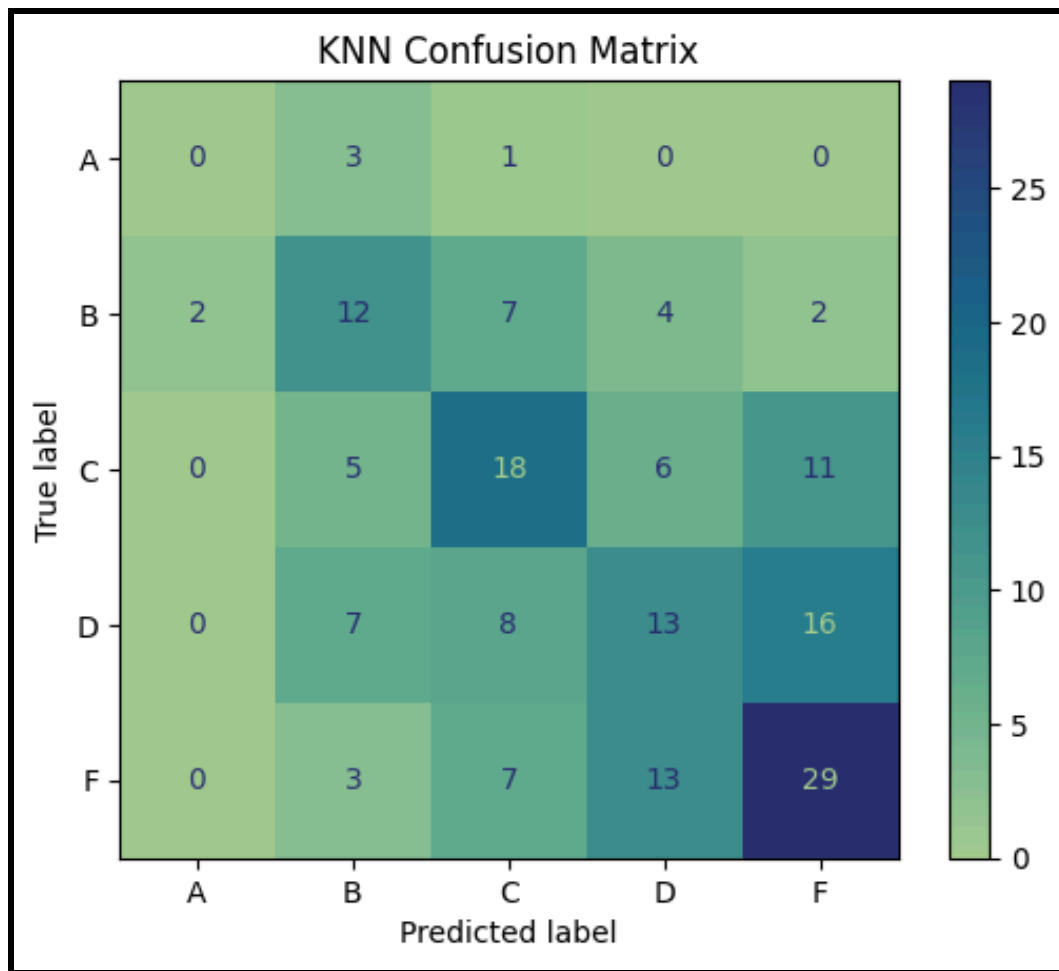


As mentioned before, the accuracies are all in the 70 percent range, with K-nearest neighbors and SVM slightly outperforming the random forest model. The SVM had a higher precision than the other two, while KNN had much higher recall than both SVM and random forest. For this problem, all the models generally struggled with falsely labeling students who passed as failing, which is shown in the lower recall across all models. Particularly, the random forest performed poorly on all metrics, likely due to having a small dataset, class imbalance, and low dimensionality. While K-nearest neighbor was the most successful out of all three models, it still had precision, recall, and accuracy under 80%. This is likely due to class imbalance and features with a lack of predictive signals for the target variable.

## Letter Grades

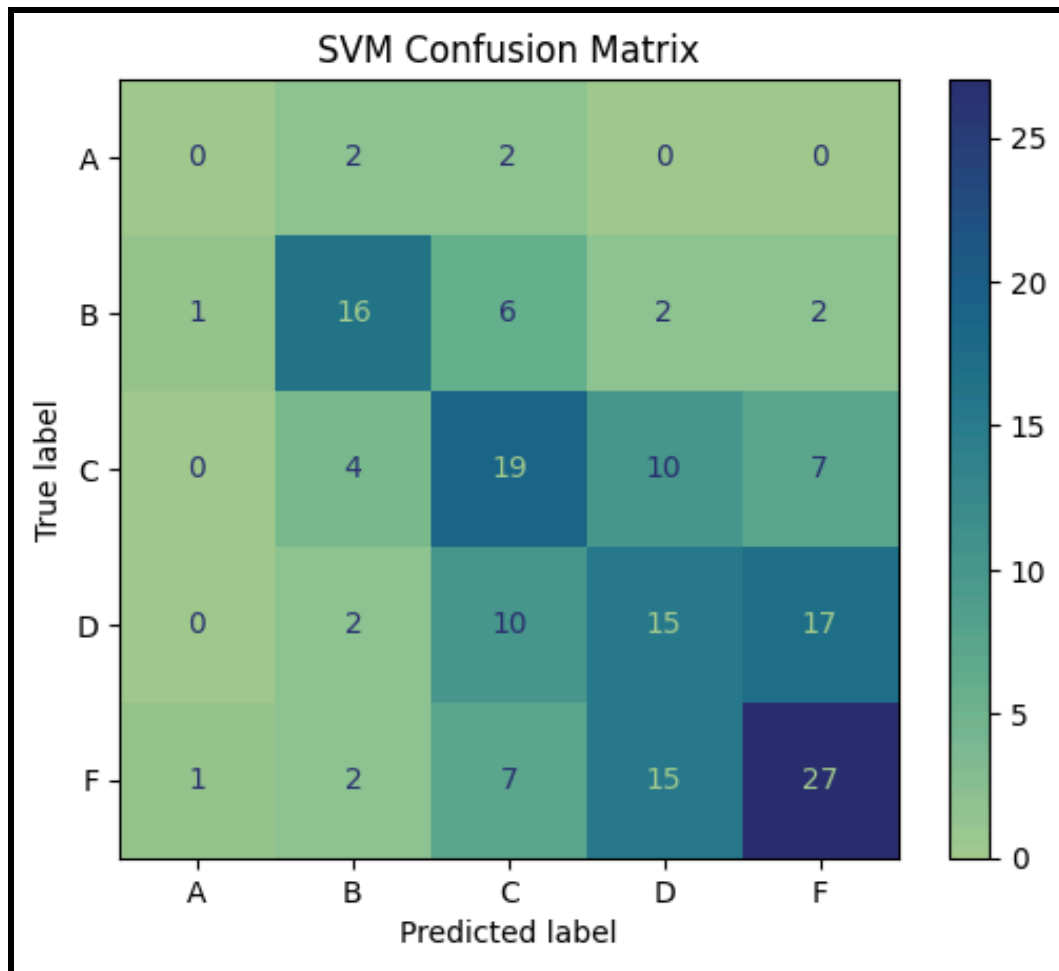
To predict a student's letter grade, K-nearest neighbors, a random forest classifier, and a support vector machine were used. The same features were used as the pass/fail section. Due to some classes being small (only 4 students got an A), five-fold cross validation was used to ensure balanced batches while still allowing for an 80/20 train test split. The performances of the models were visualized in confusion matrices.

For K-nearest neighbors, a grid search revealed that the best hyperparameters were  $K = 11$ , euclidean distance, and distance weighting. With these values, KNN got an accuracy of 43.1%, shown below:



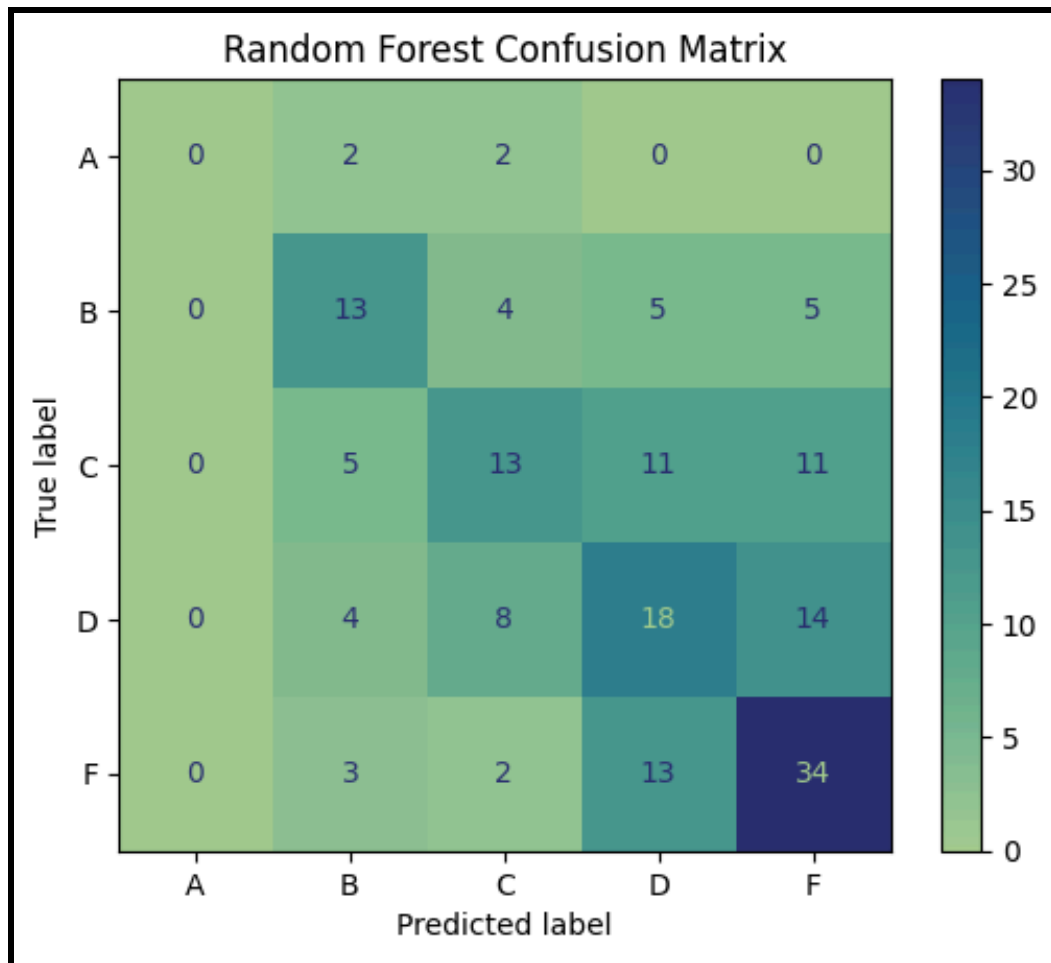
This low accuracy indicates that the KNN model was only slightly better than random guessing(20%), meaning that it struggled to learn general patterns for the letter-grade classes. Many of the model's predictions fell into a neighboring letter grade, which is likely the cause of the low accuracy. For example, students with a B often got predicted as a C or D, while students with a C or D often got predicted as F and vice versa. Out of the 4 students who got an A, none were predicted correctly. The model did best with predicting F students, as 29 of them were correct for this row, likely due to clear clusters for F students and class imbalance. This suggests that grade boundaries are fuzzy, and KNN struggled with overlapping feature values.

For SVM, the grid search revealed that the best parameters were  $C = 5$ ,  $\gamma = \text{scale}$ , and a linear kernel. With these values, the model got the following results:



The SVM achieved a 46.1% accuracy, which was a slight improvement over the 43.1% with KNN. While better, this still indicated a difficulty in separating the letter grade categories. The diagonal values in the matrix were generally higher for SVM compared to KNN, indicating it captured more of the data's structure. The model had many false values, often labeling students with a C as getting a D or F and frequently mixing up students who got a D or F. This indicates that the feature space did not form well-separated regions and that the SVM was not able to form strong boundaries. This is likely due to overlapping features between D and F students, as well as strong class imbalance, with true A and B students making up a very small amount compared to the rest.

For the random forest classifier, the model did best with 200 estimators and a max depth of 3 for the trees. With these settings, an accuracy of 46.7% was achieved, around the same as what SVM achieved.



Based on the matrix, the random forest classifier did show moderate strength on the diagonals, meaning it was able to correctly predict a good chunk of students. While it does better on predicting B students and F students correctly, the random forest still mixed up several students in the C, D, and F categories. The model was not able to pick up on the fine boundaries between letter grades, similar to the other models for this problem.

## Regression

For predicting a student's exact score, a Linear regression model with L2 (Ridge) regularization was used. For ridge regression, the alpha value with the best results was 0.1, which was found via a random search. Since ridge already penalizes coefficients to prevent overfitting, all the fields were passed into the model: prior experience, study time, readings, sleep, and predicted score, attendance, scrolling time, and leaving early.

The regression model achieved an RMSE of 8.06 and an MAE of 5.72, meaning predictions were off by about 6 points on average and up to 8 points in some cases. The  $R^2$  value of 0.416 indicates that the model explained roughly 42% of the variance in exam scores. While this shows the model captures meaningful patterns, the majority of the variation is not accounted for. This is similar to the 43-47% accuracy seen in the letter grade classifications. These results suggest that the available survey features only moderately predict exam performance, and that important

nonlinear or unobserved factors likely influence student scores. To better observe the performance, the residuals were plotted below:



The residual plot shows that the model consistently overpredicts low exam scores and underpredicts high exam scores. Instead of being randomly scattered around zero, the residuals form their own pattern, indicating that linear regression was not able to capture all relationships between the features and exam score. Additionally, the range of residuals increases for higher true scores, indicating heteroskedasticity and inconsistent performance across score ranges. These patterns support the moderate  $R^2$  value and show that the linear model is underfitting the data due to weak features. Even with having more features than the classification problems, the regression model still underperformed.

## Analysis and Conclusion

This project attempted to use student survey responses to predict exam performance through both classification and regression approaches. After preprocessing the data and dropping unimportant features, multiple algorithms were applied to predict pass/fail outcomes, letter grades, and numerical scores. Overall, the models showed limited predictive accuracy. Within the dataset, the amount of sleep a student got, their self-predicted score, and how much they had done the reading correlated the most with exam outcome. However, based on the results of the models, the

factors in the dataset alone were not enough to predict student exam outcomes with high accuracy.

Across the classification tasks, K-nearest neighbors, support vector machines, and random forests all achieved similar performance, with accuracies in the 70% range for pass/fail and only 43–47% for letter-grade prediction. The confusion matrices showed consistent patterns of misclassification, especially for neighboring grade categories, and revealed strong class imbalance and overlapping feature distributions. These results indicate that the survey features contain only moderate predictive signals for student outcomes. The regression analysis showed a similar pattern to the classification problems. The Ridge regression model achieved an RMSE of 8.06, and  $R^2$  of 0.416, indicating some predictive strength but also a large unexplained variance. Similarly, the residual plot showed signs of underfitting, with the model overpredicting low scores and underpredicting high scores.

Together, the classification and regression results suggest that exam performance is influenced by more complex patterns and other factors not captured in the dataset, such as study quality, prior academic background, or test-taking conditions. While the models identified some meaningful patterns, the limited signal in the features ultimately constrained predictive accuracy.